

Section: Particles and Waves

Topic: Spectra

The diagram shows part of the emission spectrum of a gas. It includes three visible lines of different colours.

(a)

What can be deduced about the energy levels of the atoms in the gas?

✓ The presence of discrete spectral lines shows that the **atoms have quantised energy levels**.

When electrons fall from higher to lower energy levels, they emit photons of specific energies (and therefore specific **wavelengths**), which produce the lines.

(b)

State why the gas must be at low pressure

✓ At low pressure, **collisions between atoms are infrequent**, so the emission lines are **sharp and well-defined**.

At high pressure, collisions would cause **broadening** of the lines and possible merging of spectral features.

(c)

The wavelength of one spectral line is $4.86 \times 10^{-7} \text{ m}$.

Calculate the energy of one photon of this light

Use the photon energy formula:

$$E = \frac{hc}{\lambda}$$

Where:

- $h = 6.63 \times 10^{-34} \text{ Js}$

- $c = 3.00 \times 10^8 \text{ m s}^{-1}$

- $\lambda = 4.86 \times 10^{-7} \text{ m}$

$$E = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{4.86 \times 10^{-7}} = 4.10 \times 10^{-19} \text{ J}$$

 Revision Tips

- Emission spectra = **electrons dropping to lower levels** and emitting photons

- Absorption spectra = electrons **absorbing photons** to move to higher levels

- Use:

$$E = \frac{hc}{\lambda}$$

- Low pressure gives **distinct lines** — high pressure can blur or broaden the spectrum